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RESEARCH ARTICLE

Cardiac Clarity: Harnessing Machine Learning for Accurate Heart-Disease Prediction

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ABSTRACT The major contributor to global mortality is cardiovascular disease, posing a formidable challenge to the global healthcare system. Heart disease often develops and progresses without noticeable symptoms, emphasizing the need for earlier detection to prevent severe outcomes. AI models provide tools for accurate diagnosis, data analysis, and predictive diagnosis of people who may be suffering from heart disease. But increasingly, solutions like Explainable AI (XAI) are gaining traction. XAI demystifies the decisions of AI models, making them transparent and understandable to the point where they boost confidence in the context of AI diagnostics by healthcare professionals; the AI then provides support to professionals to decide the best course of action for each patient based on the analysis. XAI techniques such as SHAP, LIME, QLatice and Anchor are popular, and have been used in this study. We analysed a heart disease dataset collected from a multispecialty hospital in India, which is publicly available on Mendeley. We used ML models, including RF, Logistic Regression, KNN, XGBoost, DT and SVM. These models were optimized and stack-ensembled to form the 'CARDIACX' model to predict the risk of heart disease. The models were fine-tuned using Grid Search, Random Search, and Bayesian Optimization methods, achieving promising results with Random Forest achieving an AUC = 0.99 and accuracy of 98.5%, outperforming all the other models and demonstrating robust performance on various metrics such as AUC, PR Curve, Log Loss, Jaccard Score, and MCC. This study provides a reliable and transparent framework for early detection of heart disease, providing useful information and enabling patients to receive timely and personalized care.

INDEX TERMS Heart disease, explainable AI, clinical evaluation, random forest, hyperparameter tuning.

I. INTRODUCTION

Cardiovascular diseases are among the leading causes of death worldwide [1]. As shown in Figure.1, The chief causes of these disorders include unhealthy food habits, a sedentary lifestyle, Smoking, unmanaged hypertension, and elevated cholesterol levels.

People may be more vulnerable to these diseases due to age and hereditary factors. Proactive measures including maintaining a nutritious diet, exercising regularly, controlling stress, and scheduling regular check-ups are essential to reducing these risks. If these preventive measures are

not adopted, it can lead to grave problems like heart attacks, strokes, and chronic illnesses that reduce life quality and put a global strain on healthcare systems [2].

Clinical evaluations, patient histories, and tests such as electrocardiograms (ECG), echocardiograms, stress tests, and blood analysis are typically used to diagnose cardiac disease. To find structural or functional heart abnormalities, advanced imaging techniques including cardiac MRIs and coronary angiography are also utilized [3]. Even though these methods are crucial in therapeutic settings, they frequently call for a large investment of time, money, and specialized knowledge.

However, there are many limitations to standard diagnostic procedures. These barriers include human-errors of clinician interpretations, ambiguous symptoms, and restricted access

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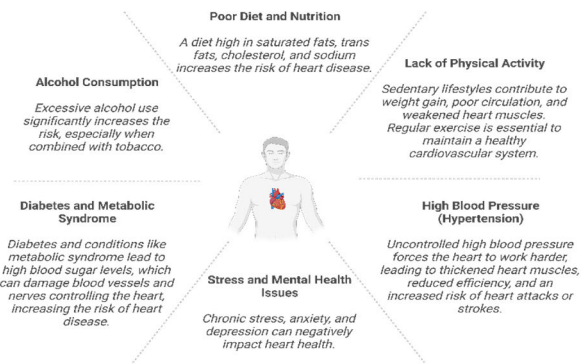


FIGURE 1. Major causes of heart disease.

to contemporary equipment in disadvantaged locations, all of which may cause diagnostic delays. Furthermore, these methods frequently fail to accurately predict future risks or the course of disease. Figure 1 visually illustrates the primary causes of heart disease [4].

One of the applications is in healthcare, where the combination of AI and ML has significantly changed the way in which healthcare works [5]. These tools examine large datasets to find patterns and connections that are difficult for people to discern. These AI-driven Models like the Random Forest classifier and Neural Networks have achieved accuracy exceeding 85% in predicting the incidence of heart diseases, thus offering scalable, efficient, and flexible solutions [6].

Explainable AI or XAI, has become an important technique for increasing transparency in AI-driven decision-making. Unlike the conventional “black-box” models, XAI provides detailed explanations of how ML algorithms reach their conclusions, allowing doctors to comprehend better, and trust the results [7]. This clarity promotes collaboration between technology and medical practices, allowing healthcare providers to implement AI in their day-to-day decision making. Within the context of cardiovascular research, XAI helps in better-informed treatments, enhances patient outcomes, and strengthens the partnership between human expertise and machine intelligence.

Many studies have been published where ML methods have been used in CVD prediction.

Ali et al. utilized ML to predict heart disease by implementing supervised learning covering RFs, Decision Trees for classification and prediction, and KNNs, resulting from this database, wherein 1,025 patient files were reported from Kaggle. Their findings indicated that the 100% accuracy reached by the RF highlights a lot about the potential benefits of machine learning in clinical applications [8].

Pires et al. used neural networks, DT, SVM and Stochastic Gradient Descent (SGD) applied to a dataset from the UCI repository, thereby obtaining a high level of 87.69% accuracy in their cross-validation experiments. Their research was also important for shedding light on feature extraction and model selection in the field of heart disease classification [9].

Moreover, Ahsan et al. through a systematic analysis, called out technical issues like data imbalance and deficiencies in interpretability for Heart Disease Diagnosis Using Machine Learning (ML). These challenges remain, notwithstanding the advances in the mixture of deep learning Models such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and others, which conspicuously improve predictive accuracy while questioning how to know why these methods yield particular results [10].

Tantimongcolwat et al. used ML to identify ischemic heart disease using magnetocardiography (MCG) data. Their study used Back-Propagation Neural Networks (BNN) and Direct Kernel Self-Organizing Maps (DK-SOM) were used to analyse MCG data from 125 participants and achieved 80.4% accuracy with DK-SOM, demonstrating the potential of using ML for non-invasive diagnostics [11].

Diwakar et al. explored image fusion and ML for the purpose of diagnosing coronary heart disease, considering such algorithms as Naïve Bayes, SVM, and ANN. According to the methods, LAW offers high precision, but long-term scalability and robustness issues related to massive data collection much remain to be worked with further [12].

The studies clearly illustrate the series of advantages that ML has sought to provide in automating as well as enhancing the accuracy of diagnostic measures for heart disease. Yet, such challenges as model interpretability, bias in the datasets, and computational costs arising from such studies still stand in the way of designs for deep automation for free screening.

There were several research gaps in the existing studies on heart disease prediction as mentioned in Table 6. The majority of researchers did not employ the Explainable AI (XAI) techniques to make their models interpretable. Work had been done where XAI techniques were utilized, but restriction to SHAP (SHapley Additive exPlanations) is largely observed. Different XAI techniques can be applied to make the output of models more transparent and interpretable. This is vital for establishing confidence from medical professionals as well as patients. Further, several heart disease Classifiers were not optimized through hyperparameter tuning and feature selection, which are crucial for improving performance of the model. This study addresses the limitations of traditional diagnostic approaches—such as, limited interpretability, and model optimization by combining robust machine learning classifiers with diverse explainable AI techniques like SHAP, LIME, QLattice and Anchor, to develop an interpretable and generalizable heart disease prediction framework.

The work done in this research are:

1. Three hyperparameter tuning methods were employed to optimize the classifiers: (a) Bayesian Optimization, (b) Grid Search and (c) Random Search.
2. All individual classifiers were stack-ensembled using a customized approach to create the “CARDIACX” model, as shown in Figure 2.
3. Four XAI techniques were utilized to ensure the predictions were interpretable and transparent: (a) SHAP, (b) LIME, (c) QLattice and (d) Anchor.

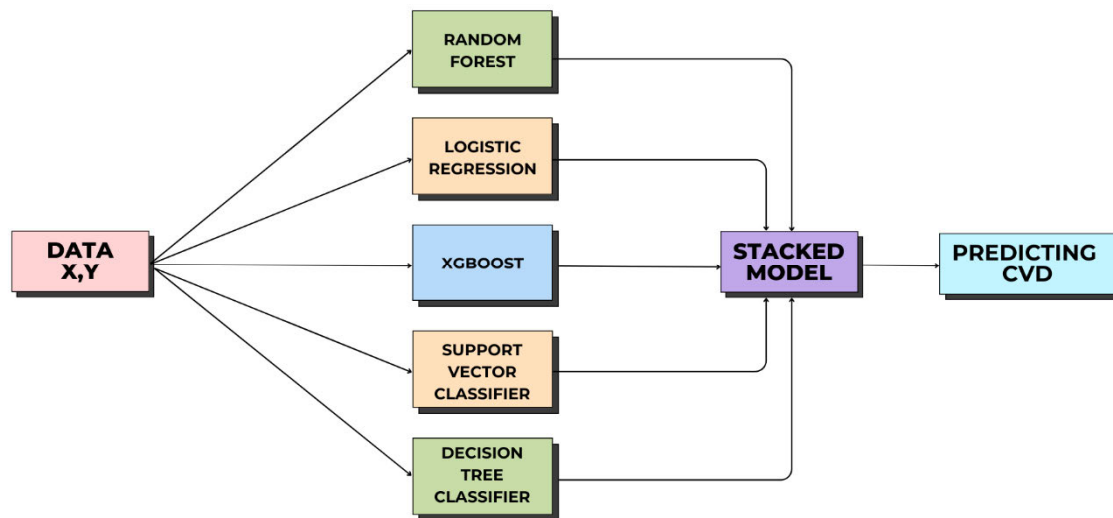


FIGURE 2. Architecture of the CARDIACX stack ensemble model using 5 baseline- classifiers.

4. The key markers highlighted by the explainers were thoroughly examined from a medical perspective for further validation.

II. MATERIALS AND METHODS

A. DATASET

The cardiology dataset that has been used is obtained from a multispecialty hospital in India and is freely available on Mendeley. It contains over 14 common features, making it one of the most comprehensive cardiology datasets available for research. The dataset includes 1,000 subjects with 12 characteristics [13]. All patient records in the dataset were fully anonymized prior to publication, with no personally identifiable information (PII) or sensitive health identifiers included. There were no missing data fields in the dataset. This dataset has been useful in developing early detection of cardiac diseases and generating predictive machine learning models.

B. DATA PREPROCESSING

The first step of data-processing was removing features such as ‘patientid’ and the target variable ‘target’. The dataset did not have any null values. Numerical features such as ‘age’, ‘restingBP’, ‘serumcholesterol’, ‘maxheartrate’ and ‘oldpeak’ had to undergo normalization.

C. METHODOLOGY

The machine learning pipeline used in this study is shown in Figure 3. The descriptive statistical parameters such as mean, median, SD, IQR and Range for the continuous attributes has been tabulated in Table 1. Violin plots have also been illustrated in Fig. 3 for identifying variation in markers. The mean age was found to be almost the same in patients with and without heart disease. The following plots are for ‘age’, ‘restingBP’, ‘serumcholesterol’, ‘maxheartrate’ and ‘oldpeak’.

Inferential statistical tests, including t-tests and chi-square tests, were performed [14]. The t-test used in this study is Student’s t-test. p value less than 0.001 indicates that the attribute is deemed significant for heart disease prediction. The results of the t-tests have been shown in Table 2. Features like ‘restingBP’, ‘serumcholesterol’ and ‘maxheartrate’ are important. Attributes like ‘chestpain’, ‘fastingbloodsugar’, ‘restingelectro’, ‘noofmajorvessels’, and ‘slope’ were significant. The Descriptive statistical measures for categorical attributes [15] have been laid out in Table 3. The violin plots and bar plots for all the categorical attributes are illustrated in Figure 4 and Figure 5 respectively. Table 4 has the chi square tests for the same attributes. There is a strong relation between slope and target. Categories 2 and 3 of “slope” appear to be strongly associated with heart disease (target = 1), while category 0 is almost exclusively associated with the absence of the disease. Although only 5 attributes have a p value less than 0.001, no attributes were excluded for the model input as they were significant features.

The dataset contained no missing values. To prevent potential biases, data has been normalized using StandardScaler() as many ML algorithms perform better when the features are standardized to zero mean and unit variance. This method also handles outliers, better than min-max scaling in certain clinical datasets [16]. The dataset had no target class imbalance [17]. After initial data exploration, it was divided into training and testing subsets, with an 80% training and 20% testing split [18].

In this study, we used a stacking method to assemble several classifiers [19]. Stacking enhances model performance by integrating the predictive capabilities of multiple classifiers through a meta-learner. This approach is beneficial as it can capture diverse patterns in data that individual algorithms might overlook. Additionally, combining different models helps reduce overfitting, particularly when prediction bias

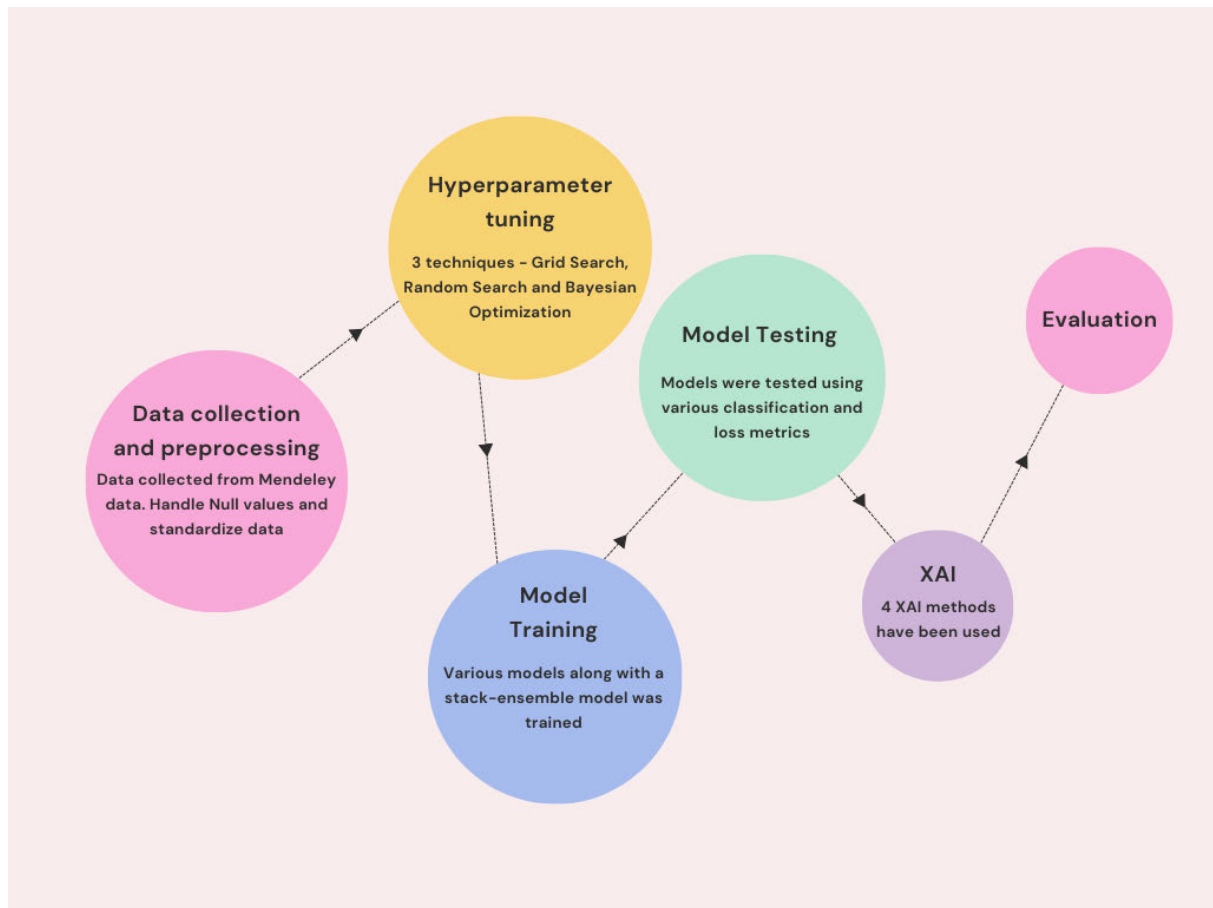


FIGURE 3. ML pipeline used in this study to detect CVD.

or errors differ across various data subsets [20]. We have used 4 XAI methods to make the machine learning algorithm interpretable.

1) SHAP

SHAP (SHapley Additive exPlanations) is a powerful and commonly used technique for interpreting ML models [21]. It is derived from the Shapley value in cooperative game theory, which equitably allocates model outputs among input features, explaining their contribution to the prediction. SHAP provides a unified framework for feature mapping, and thus lends itself to both global and local interpretations. While local explanations help understand individual predictions, global explanations provide insight into overall model behaviour. The main advantages of SHAP are its consistency and theoretical guarantees, ensuring that more important features receive higher attribute values. SHAP values can be visualized using summary graphs, dependency graphs, and decision diagrams, Enhancing the interpretability of complex models like neural networks and ensemble methods. SHAP is model-independent, meaning that it can be applied to any model. Certain implementations, such as TreeSHAP, are Designed for optimization in tree-based models like Random

Forest and XGBoost, but it can work with any machine learning algorithm.

2) LIME

LIME is an explainability technique that helps interpret machine learning models by generating locally faithful approximations [22]. It perturbs the input data and trains simple, interpretable models (such as LR or DT) to estimate the predictions of a black-box model within the context of a specific instance. This allows users to understand which features contribute most to a particular prediction. LIME is model-agnostic, allowing it to interpret any machine learning algorithm, which is why it is widely used in applications such as healthcare and finance. However, the system also has certain limitations that include very high sensitivity to disturbances and instabilities for similar inputs when written in different ways.

3) QLATTICE

QLattice is an AI-powered symbolic regression method developed by Abzu that helps discover mathematical relationships in data [23]. Unlike traditional machine learning models, QLattice focuses on generating interpretable models

TABLE 1. Descriptive statistics for continuous variables.

	target	N	Missing	Mean	Median	SD	Variance	Minimum	Maximum
age	Absence of heart disease	420	0	49.07	49.00	18.70	349.86	20	80
	Presence of heart disease	580	0	49.37	49.00	17.25	297.44	20	80
restingBP	Absence of heart disease	420	0	134.77	130.00	26.56	705.47	94	200
	Presence of heart disease	580	0	164.04	168.00	26.04	678.22	94	200
serumcholesterol	Absence of heart disease	420	0	281.06	270.00	87.56	7666.81	132	465
	Presence of heart disease	580	0	333.45	351.50	153.50	23562.62	0	602
maxheartrate	Absence of heart disease	420	0	136.31	134.00	39.26	1541.28	71	202
	Presence of heart disease	580	0	152.12	152.00	28.22	796.4	96	202
oldpeak	Absence of heart disease	420	0	2.51	2.30	1.72	2.95	0	6.20
	Presence of heart disease	580	0	2.85	2.70	1.71	2.92	0	6.20

by identifying simple and elegant equations that describe the relationships between features and outcomes. It is particularly useful for scientific research as it avoids overfitting by prioritizing accurate and explainable models. QLattice efficiently explores complex data sets while ensuring that discovered patterns are human-readable, making it a valuable tool in fields such as genetics, physics, and financial modelling.

4) ANCHOR

Anchor is an explainability technique that provides highly accurate, human-interpretable rules to explain individual predictions [24]. Unlike LIME, which perturbs input functions to approximate local decision boundaries, Anchor creates reliable “if-then” decision rules that act as explanations. These rules act as “anchors” to ensure that similar inputs receive the same predictions. Anchor is particularly effective for high-dimensional data and complex models, as it provides more stable and consistent explanations compared to LIME. However, it can run into problems with feature interactions or when simple rules do not exist.

Hyperparameters are adjustable settings that influence the behaviour of a machine learning algorithm before training begins [25]. Hyperparameters significantly affect model training by influencing training time, resource requirements (and associated costs), model convergence, and accuracy. In this study, three hyperparameter tuning techniques were employed:

5) GRID SEARCH

Grid Search sets up a hyperparameter grids and their different values [26]. For each possible combination the model is trained and evaluated on validation data, systematically testing all possible combinations of the specified hyperparameter values. This approach exhaustively searches all possible combinations, which can be very inefficient in terms of training time and cost.

6) RANDOM SEARCH

Random Search operates similarly to Grid Search but selects hyperparameter combinations randomly instead of

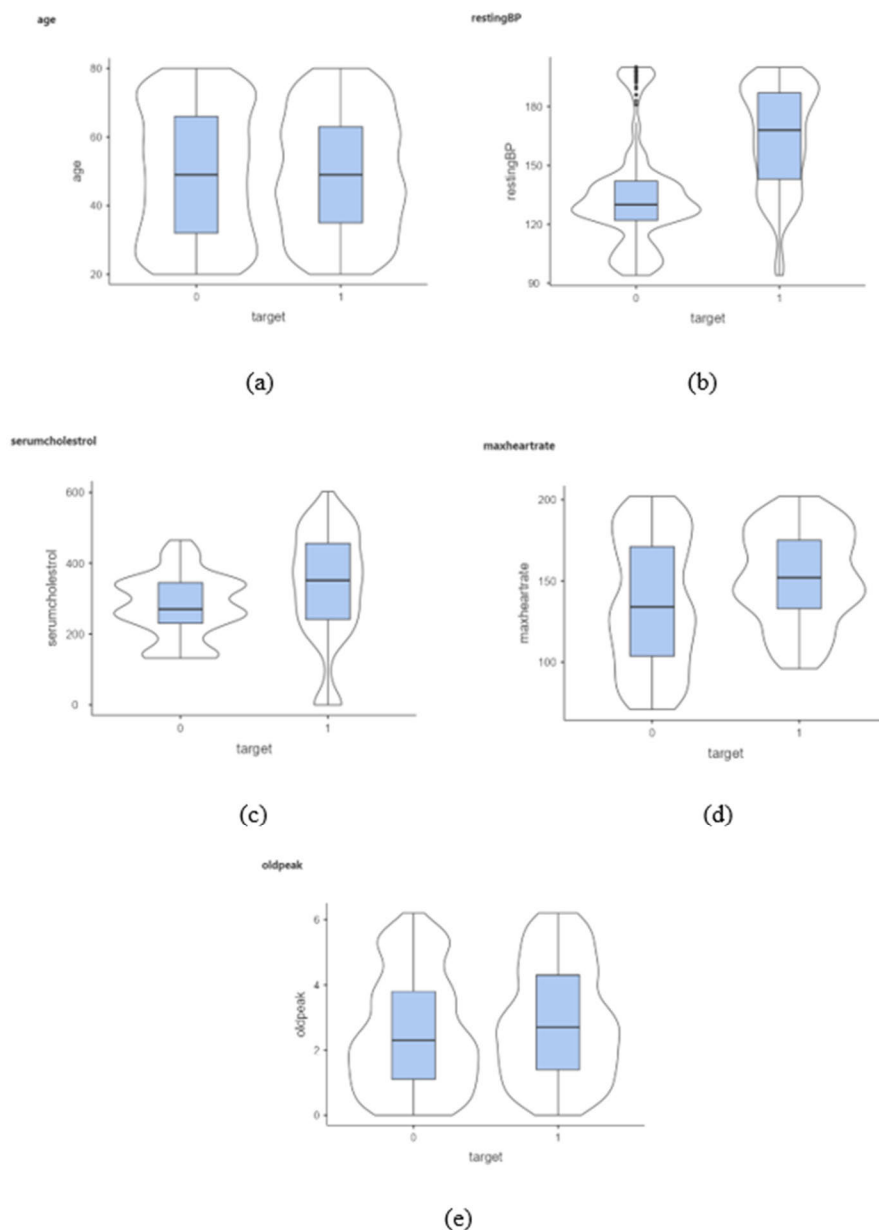


FIGURE 4. Violin plots for analyzing marker variation. (a) age (b) restingBP (c) serumcholesterol (d) maxheartrate (e) oldpeak.

TABLE 2. Independent T-Test (student's t) samples.

	Statistic	df	p
age	-0.264	998	0.792
restingBP	-17.397	998	< 0.001
serumcholesterol	-6.292	998	< 0.001
maxheartrate	-7.409	998	< 0.001
oldpeak	-3.113	998	0.002

exhaustively testing all possibilities [27]. The number of search iterations can be adjusted according to time and resource limitations.

7) BAYESIAN OPTIMIZATION

Bayesian Optimization is a method that uses Bayesian statistics to efficiently Explore the range of possible

TABLE 3. Descriptive statistical measures for categorical attributes.

	target	N	Missing	Mean	Median	SD	Variance	Minimum	Maximum
gender	Absence of heart disease	420	0	0.757	1.00	0.429	0.184	0	1
	Presence of heart disease	580	0	0.771	1.00	0.421	0.177	0	1
chestpain	Absence of heart disease	420	0	0.360	0.00	0.678	0.46	0	3
	Presence of heart disease	580	0	1.429	2.00	0.868	0.753	0	3
fastingbloodsugar	Absence of heart disease	420	0	0.133	0.00	0.340	0.116	0	1
	Presence of heart disease	580	0	0.414	0.00	0.493	0.243	0	1
restingrelectro	Absence of heart disease	420	0	0.362	0.00	0.519	0.27	0	2
	Presence of heart disease	580	0	1.028	1.00	0.801	0.642	0	2
exerciseangia	Absence of heart disease	420	0	0.521	1.00	0.500	0.25	0	1
	Presence of heart disease	580	0	0.481	0.00	0.500	0.25	0	1
slope	Absence of heart disease	420	0	0.600	1.00	0.546	0.298	0	2
	Presence of heart disease	580	0	2.221	2.00	0.646	0.418	1	3
noofmajorvessels	Absence of heart disease	420	0	0.660	0.00	0.827	0.683	0	3
	Presence of heart disease	580	0	1.629	2.00	0.871	0.759	0	3

hyperparameter values for a machine learning model [28]. The search is iteratively adjusted based on previous evaluations. The goal is to find the optimal combination of hyperparameters much faster than traditional grid or random search methods, through an intelligent balance of exploration and exploitation.

III. RESULTS

This study adopts three approaches for finding optimal hyperparameters. The results have been shown in Table 7,

along with the hyperparameters for each algorithm in Table 8. For the Random Forest model, the number of estimators was varied between 100 and 1000, and max_depth was explored within the range of 10 to 150 to balance bias-variance trade-offs. Parameters such as min_samples_split and min_samples_leaf were adjusted to avoid overfitting. Similarly, for Logistic Regression, the regularization parameter C was tuned across a logarithmic scale from 0.001 to 1000. The chosen ranges aimed to cover both underfitting and overfitting regions. Grid Search with 5-fold

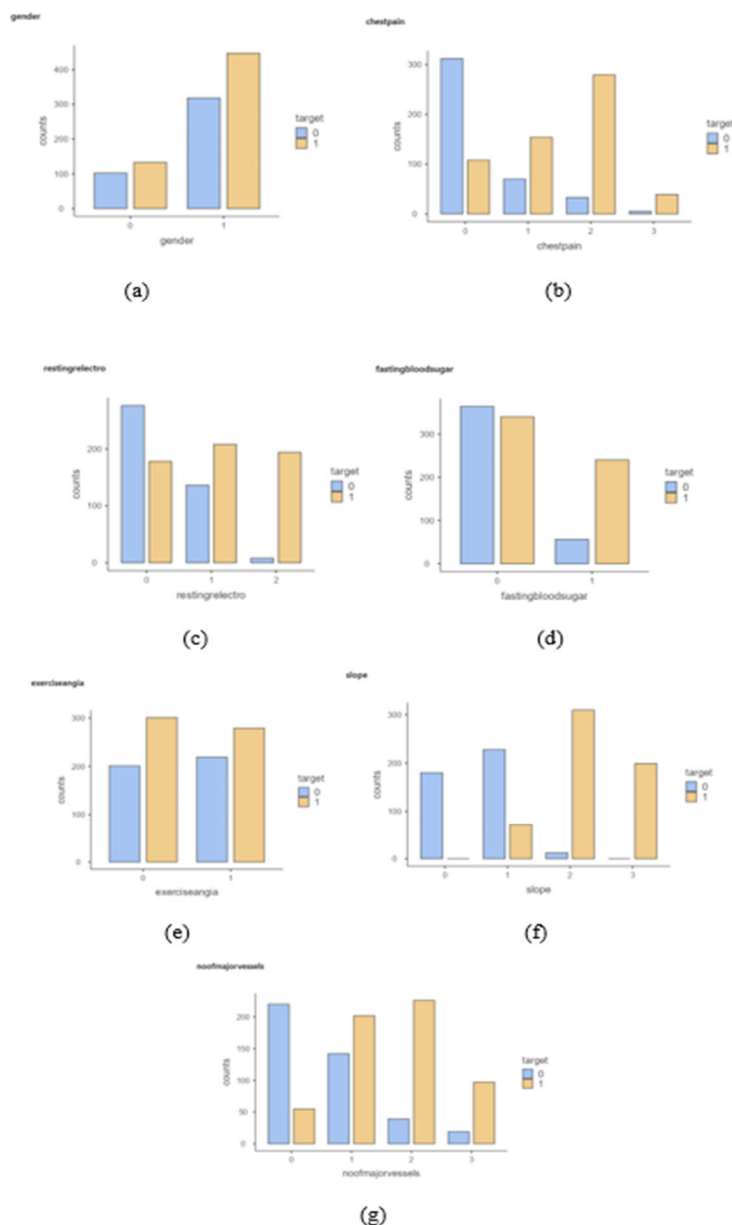


FIGURE 5. Bar Plots as part of statistical analysis.

TABLE 4. Chi square tests for categorical attributes (student's t).

Attribute	df	p
chestpain	3	< 0.001
fastingbloodsugar	1	< 0.001
restingelectro	2	< 0.001
noofmajorvessels	3	< 0.001
slope	3	< 0.001

cross-validation was used where computation was feasible, along with Random Search and Bayesian Optimization. When grid search was used, RF, KNN and CARDIACX

got an accuracy of 97%. The XGBoost model achieved 99% accuracy. A 97% F1 score was obtained with the Random Forest method using Grid Search. For random search,

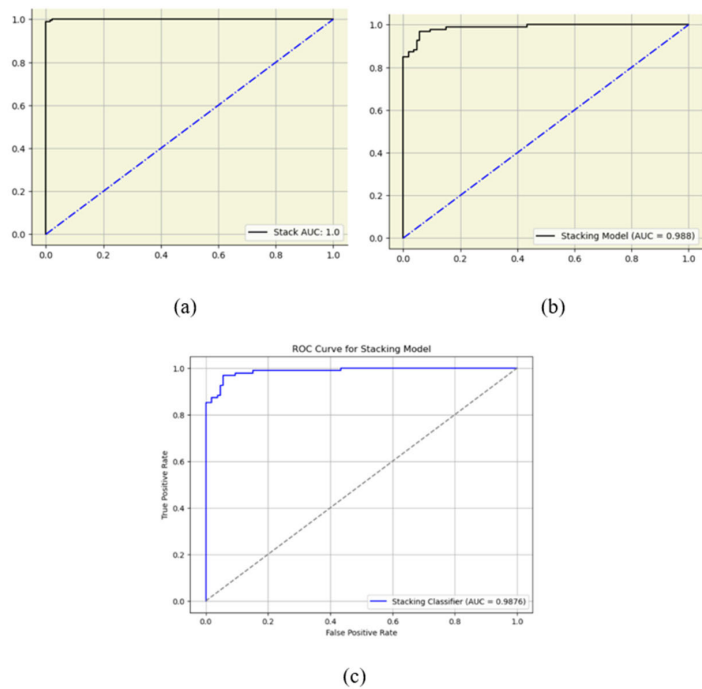


FIGURE 6. AUC curve for the CARDIACX model (a) Grid Search, (b) Random Search and (c) Bayesian Search.

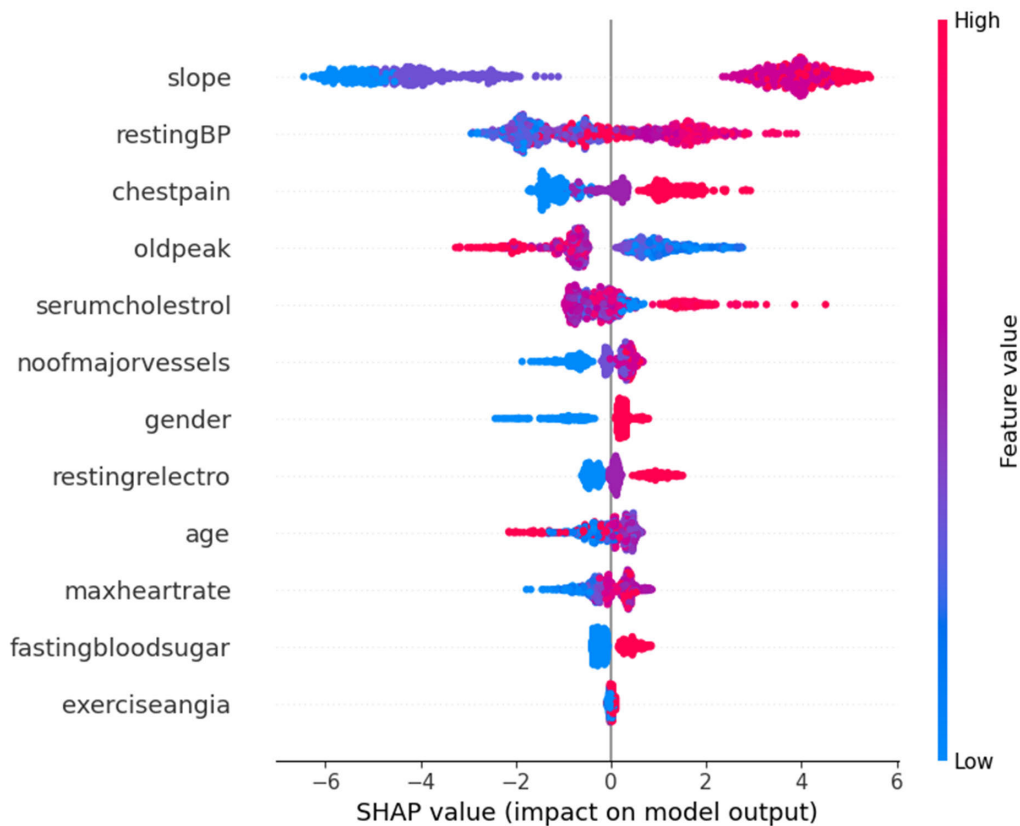


FIGURE 7. SHAP bee swarm plot for a heart patient.

random forest achieved 98.5% accuracy and CARDIACX achieved 98%. XGBoost achieved 97.45% F1 score when

using random search. For the Bayesian optimization; CARDIACX model achieved 98% accuracy. XGBoost achieved an



FIGURE 8. SHAP force-plot for a heart patient.

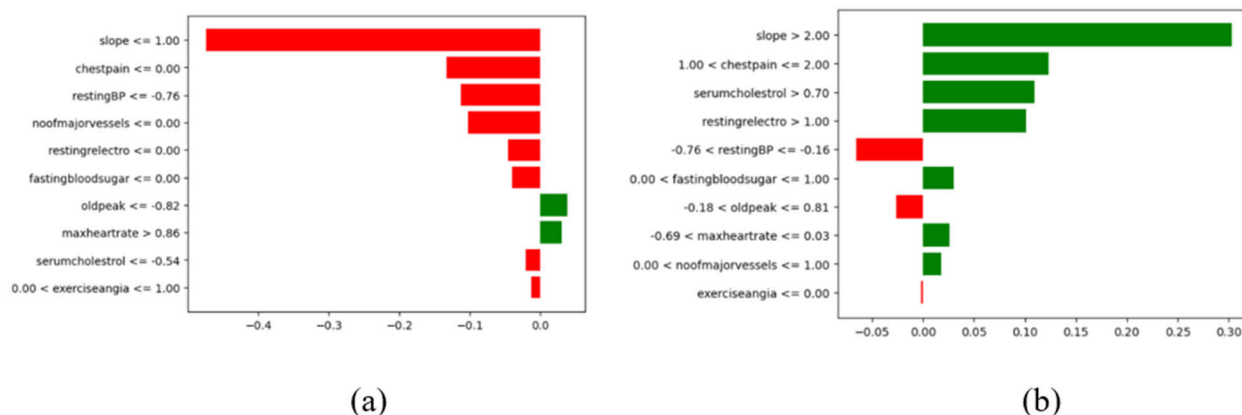
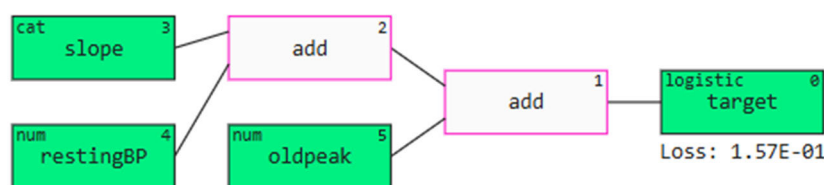


FIGURE 9. LIME plots to analyse prediction (a) non-CVD diagnosis (b) CVD diagnosis.



Epoch no. 10/10 - Tried 8381 models - Completed in 25s.

FIGURE 10. QGraph to identify important markers in heart disease prediction.

accuracy of 97.5%. The AUC curves for the ensemble model is illustrated in Fig.6.

Additionally, there were four explainers used to elaborate on the predictions, and further analysis was conducted using the CARDIACX model. Fig.7 shows the bee-swarm plot for the SHAP explainer. The markers are arranged in descending order with respect to significance. These were – slope (A steep slope during peak exercise is often associated with myocardial ischemia, making it a critical indicator in stress testing and heart disease diagnosis), restingBP (Elevated resting systolic pressure is a well-known risk factor for hypertension and cardiovascular complications), chestpain (This feature differentiates between typical angina, atypical angina, and non-anginal pain, which are routinely used in cardiology for risk stratification), oldpeak (A greater degree of ST depression typically indicates more severe cardiac stress or oxygen deprivation in heart tissues) and serum-cholesterol (Elevated levels of LDL cholesterol contribute

to atherosclerosis, increasing the risk of coronary artery blockage and heart attacks). In addition, there is a vertical plane dividing the two classes. Blue means low, and red means high. The higher the slope, the greater the chance for diagnosis of heart disease. If restingBP is high, the likelihood of heart disease increases. SHAP provides local interpretability for individual predictions using a force plot, as shown in Fig. 8. LIME predictions for a heart disease patient are illustrated in Fig. 9(b), where attributes such as slope, chest pain, serum cholesterol, resting electrocardiogram, fasting blood sugar, and maximum heart rate all indicate the same outcome. Similarly, Fig. 9(a) presents LIME predictions for a non-heart disease patient. A QGraph generated from the QLattice model is depicted in Fig. 10, highlighting restingBP, oldpeak, and slope as the most distinguishing markers. In this study, the QLattice model employed the “addition” activation function. The final explainer used is Anchor, which provides explanations based on specific conditions evaluated by

TABLE 5. Anchor explanations.

PATIENT TYPE	ANCHOR	PRECISION	COVERAGE
Non-CVD	slope ≤ 1.00 AND max heartrate ≤ -0.72 AND restingrelectro ≤ 1.00 AND noofmajorvessels ≤ 1.00 AND oldpeak > -0.88 AND $-0.01 < \text{age} \leq 0.83$ AND chestpain ≤ 2.00	0.95	0.02
Non-CVD	slope ≤ 1.00 AND chestpain ≤ 0.00 AND restingBP ≤ -0.77	1.0	0.15
Non-CVD	slope ≤ 1.00 AND chestpain ≤ 0.00 AND restingBP ≤ -0.16	0.97	0.25
Non-CVD	slope ≤ 1.00 AND noofmajorvessels ≤ 0.00 AND restingBP ≤ -0.77	0.95	0.09
Non-CVD	slope ≤ 1.00 AND chestpain ≤ 0.00 AND restingBP ≤ -0.16	0.98	0.27
CVD	slope > 2.00 AND chestpain > 2.00	1.00	0.01
CVD	slope > 1.00 AND chestpain > 1.00	1.00	0.29
CVD	slope > 2.00 AND restingrelectro > 1.00	1.00	0.07
CVD	slope > 2.00 AND chestpain > 2.00	1.00	0.01
CVD	slope > 2.00 AND chestpain > 1.00	1.00	0.12

TABLE 6. A few studies that used AI for predictions.

Author	Dataset Size	ML models used	Maximum results	XAI
Gonsalves et al.	462	Naïve Bayes, Support Vector Machine (SVM), and Decision Tree.	Accuracy: 72.3% (Naive Bayes)	-
Nikhar & Karandikar	303	Decision Tree, Naïve Bayes	Accuracy: 74.2% (Decision Tree)	-
Anusha et al.	303	Random Forest, Logistic Regression	Accuracy: 85.81% (Random Forest)	-
This Study	1000	Various models + CARDIACX (customized ensemble model)	Accuracy: 98%	SHAP, LIME, QLattice, Anchor

precision (correctness) and coverage (scope). Table 5 presents Anchor-based explanations for heart disease-positive and negative patients. Slope, restingBP, chestpain, restingrelectro are some important ones. Four XAI methods have been adopted and, in their opinion, the most important variables for the identification of heart diseases are slope, chestpain, restingBP, restingrelectro, and oldpeak. These can be used to predict cardiovascular diseases in a patient.

IV. DISCUSSION

This study employed multiple classification methods to diagnose cardiovascular diseases. To enhance performance, three hyperparameter tuning techniques were applied, with Random Search proving to be the most effective. The custom stack algorithm, “CARDIACX,” achieved an accuracy of 98%. Additionally, four XAI techniques were used to identify key markers. Important attributes are slope, chestpain, restingBP, restingrelectro, and oldpeak.

In this study, the higher the ST segment slope during exercise, the Greater likelihood of a heart disease diagnosis. If the old peak (representing the difference in ST segment observed during exercise compared to rest, as measured by electrocardiogram) was high, the risk of heart disease was very low. Chest pain was observed in patients with and without heart disease. Resting blood pressure was higher in patients with heart disease. In this study, serum cholesterol levels were higher in patients with heart disease. Finally, age also plays an important role. Younger age groups are less likely to have heart disease. These trends have also been reported in similar heart disease studies [29]. The variability of these signs, accurately detected by classifiers, can aid physicians and healthcare professionals in making precise diagnoses. Numerous researchers have applied machine learning (ML) for heart disease diagnosis. Gonsalves et al. employed machine learning

techniques to predict coronary heart disease (CHD) using a dataset of 462 cases, where Naïve Bayes achieved the highest accuracy of 72.3% [30]. Similarly, Nikhar and Karandikar explored machine learning methods and found that decision trees outperformed Naïve Bayes, achieving a maximum accuracy of 74.2% [31]. Anusha et al. utilized Random Forest and Logistic Regression to evaluate heart disease risk, among which random forest achieved the highest accuracy of 85.81% [32].

Mahmoud et al. [33] demonstrates the use of physical and exercise parameters for heart rate modelling, further underscoring the value of incorporating physiological insights into ML-based cardiovascular diagnostics.

Table shows the comparison between different models with our study. Unlike previous studies, our study incorporates five explainable artificial intelligence (XAI) techniques to enhance interpretability in heart disease prediction. All of the above studies are tabulated in Table 6.

This study has some limitations. The dataset used in this study only contains 1000 cases from a single multi-speciality hospital in India, which limits the generalizability of the result to a broader and more diverse population. The lack of demographic diversity including limited representation across age groups, gender, ethnic backgrounds, and geographical regions, could impact the robustness of the model when applied to other clinical settings. Both SHAP and LIME can be computationally intensive, especially when applied to complex ensemble models. SHAP requires multiple model evaluations to compute Shapley values, which can significantly increase processing time. LIME’s local explanations can be sensitive to the choice of parameters (e.g., kernel width) and may produce unstable or inconsistent results across runs. Both techniques also assume feature independence when perturbing input variables, which might not hold true in real-world clinical data where features are often correlated. Additionally, deep learning

TABLE 7. Results of hyperparameter tuning.

GRID SEARCH									
Algorithm	Accuracy	Precision	Recall	F1-score	AUC	Hamming loss	Jaccard score	Log loss	Matthew's correlation coefficient
Random Forest	97	97	97	97	1.0	0.03	0.95	1.081	0.938
Logistic Regression	96	96	97	97	1.0	0.04	0.934	1.441	0.917
Decision Tree	92	95	91	93	0.96	0.08	0.869	2.883	0.837
KNN	97	99	96	97	1.0	0.03	0.949	1.081	0.939
XGBoost	99	99	100	100	1.0	0.005	0.991	0.18	0.989
SVC	96	96	96	96	0.98	0.041	0.931	1.484	0.91
CARDIACX	97	98	97	98	1.0	0.025	0.957	0.041	0.948

RANDOM SEARCH									
Algorithm	Accuracy	Precision	Recall	F1-score	AUC	Hamming loss	Jaccard score	Log loss	Matthew's correlation coefficient
Random Forest	98.5	99.13	98.29	98.71	0.99	0.015	0.974	0.081	0.969
Logistic Regression	97	97.43	97.43	97.43	0.99	0.03	0.95	0.085	0.938
Decision Tree	97	95.86	99.14	97.47	0.96	0.03	0.95	1.081	0.938
KNN	96	97.39	95.72	96.55	0.99	0.04	0.933	0.079	0.918
XGBoost	97	96.63	98.29	97.45	0.99	0.03	0.95	0.053	0.938
SVC	98	97.47	99.14	98.3	0.99	0.02	0.96	0.069	0.958
CARDIACX	98	97.47	99.14	98.3	0.99	0.02	0.96	0.054	0.958

BAYESIAN OPTIMIZATION									
Algorithm	Accuracy	Precision	Recall	F1-score	AUC	Hamming loss	Jaccard score	Log loss	Matthew's correlation coefficient
Random Forest	98	98.29	98.29	98.29	0.99	0.02	0.966	0.076	0.958
Logistic Regression	96	96.58	96.58	96.58	0.99	0.04	0.933	0.081	0.917
Decision Tree	96.5	95.83	98.29	97.04	0.96	0.03	0.942	1.09	0.927
KNN	97	97.43	97.43	97.43	0.99	0.03	0.95	0.085	0.938
XGBoost	97.5	98.27	97.43	97.85	0.99	0.02	0.957	0.05	0.948
SVC	98	97.47	99.14	98.3	0.99	0.02	0.96	0.069	0.958
CARDIACX	98	97.47	99.14	98.3	0.99	0.02	0.96	0.05	0.958

models remain underexplored because they rely on large-scale data sets. Cloud-based infrastructure, which could

improve data accessibility and security, was not deployed in this study.

TABLE 8. Hyperparameters for each algorithm.

Algorithm	Hyperparameters
Random Forest	{'bootstrap': True, 'max_depth': 100, 'max_features': 3, 'min_samples_leaf': 3, 'min_samples_split': 10, 'n_estimators': 200}
Logistic Regression	{'C': 100, 'penalty': 'l2'}
Decision Tree	{'criterion': 'gini', 'max_depth': 120, 'max_features': 'log2', 'min_samples_leaf': 1, 'min_samples_split': 10, 'splitter': 'best'}
XGBoost	{'colsample_bytree': 0.3, 'gamma': 0.1, 'learning_rate': 0.15, 'max_depth': 4, 'min_child_weight': 1}
KNN	{'n_neighbors': 10}
SVC	{'C': 100, 'coef0': 0.0, 'degree': 2, 'gamma': 'scale', 'kernel': 'rbf'}
CARDIACX	Best parameters: {'final_estimator__C': 10, 'final_estimator__max_iter': 100}

V. CONCLUSION

XAI algorithms were employed to interpret the predictions of the customized ensemble model. The dataset was initially analyzed using statistical tests to identify key insights. Three hyperparameter optimization techniques were utilized to fine-tune the models, including Bayesian Optimization, Randomized Search, Grid Search. A custom model was developed with the addition of multiple classifiers to form a 'custom ensemble model'. All three optimization methods were applied for training and testing the models, with Bayesian Optimization achieving an AUC of 0.99. Four explainers, including SHAP and LIME, provided evidence that the model's predictions had an impact on clinical variables. The risk of CVD was significantly increased when analyzing blood pressure, serum-cholesterol and other factors. XAI techniques are utilized to assist doctors in comprehending the assumptions used in model predictions, leading to improved patient outcomes and diagnostic precision. This study primarily focuses on supervised learning. Future research could explore federated learning, unsupervised learning, and reinforcement learning to enhance

predictive capabilities. Additionally, a user-friendly interface could be developed for real-time clinical diagnosis. Expanding the dataset with diverse patient records from various healthcare facilities would further improve model reliability and generalizability.

Several recent studies have demonstrated the effectiveness of artificial intelligence and deep learning across domains beyond cardiovascular research. For instance, deep learning has been applied for sentiment classification, network intrusion detection, and behavioural modelling [34], [35], [36]. These applications underscore the versatility and scalability of AI, validating its potential in health informatics as well.

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